

Notes API

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1. PROJECT TITLE AND PROBLEM STATEMENT

Project Title

Simple Notes API using Docker and GitHub Actions

Problem Statement

In modern software systems, managing and accessing data through APIs is essential. However, many applications rely heavily on frameworks, making it difficult to understand the core working of HTTP servers and request handling.

The objective of this project is to develop a simple Notes API using pure Java without any external frameworks. The system allows users to add, retrieve, and delete notes through HTTP requests.

Additionally, the project demonstrates the use of Docker for containerization and GitHub Actions for continuous integration, ensuring portability and automation in the development process.

2. TOOLS / TECHNOLOGIES USED

Technologies Used: Docker + GitHub Actions

1. Java (Core Java)

- Used to implement the application logic
- Utilizes built-in HTTP server ([HttpServer](#))
- Handles GET, POST, and DELETE requests

2. Docker

- Used to containerize the application
- Ensures the application runs consistently across different environments
- Eliminates dependency issues

3. GitHub

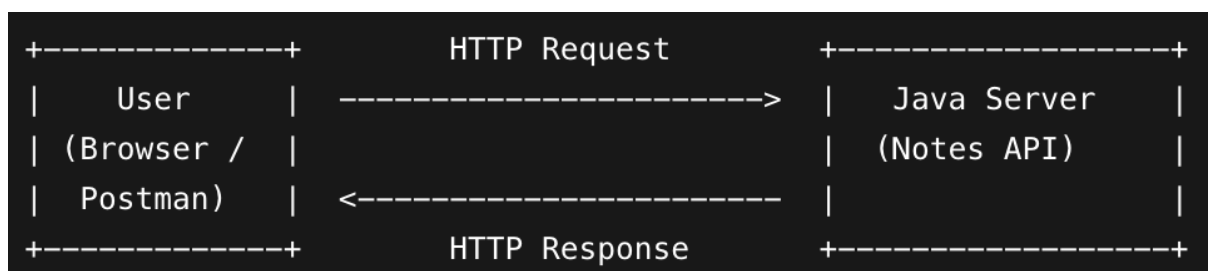
- Used for version control
- Stores and manages project code

4. GitHub Actions

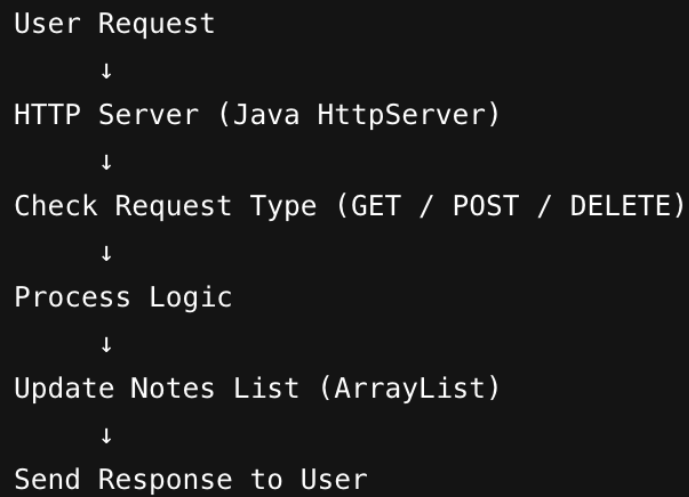
- Used for Continuous Integration (CI)
- Automatically compiles the project on every code push
- Detects errors early in the development cycle

3. ARCHITECTURE / WORKFLOW DIAGRAM (CI/CD PIPELINE)

1. System Architecture Diagram



2. API Request Flow Diagram



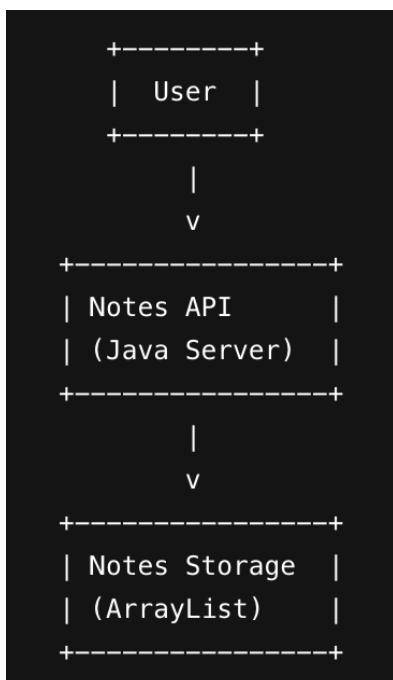
3. CI/CD Workflow Diagram



4. Docker Workflow Diagram



5. Data Flow Diagram



4.EXPECTED RESULTS AND CONCLUSION

Expected Results

- The application successfully handles HTTP requests
- Users can:
 - Add notes
 - Retrieve notes
 - Delete notes
- The application runs inside a Docker container
- GitHub Actions automatically builds the project

Conclusion

This project demonstrates the development of a simple Notes API using pure Java without relying on frameworks. It highlights the core concepts of HTTP request handling and server implementation.

By integrating Docker, the application achieves platform independence and consistent execution across environments. The use of GitHub Actions introduces automation through continuous integration, improving efficiency and reliability.

Overall, the project provides a clear understanding of backend development fundamentals along with modern DevOps practices.